



AWS GenAI IDP Accelerator

Since its release in 2025, Amazon Web Services' GenAI IDP Accelerator has been regularly updated with an impressive array of functionality that makes it more user-friendly to non-developers while retaining its deep configurability and sophistication. It has seen a growing base of customer engagements, with claims processing established as a flagship use case.

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The Company

Amazon Web Services (AWS) introduced the GenAI IDP Accelerator in 2025 as an open-source framework to streamline intelligent document processing (IDP) projects. Developed by the AWS Generative AI Innovation Center (GenAIIC), the accelerator is designed to reduce time-to-value for customers by offering prebuilt patterns, scalable infrastructure, and integration with foundational AI services. It reflects AWS's broader strategy to democratize access to generative AI and agentic automation while maintaining enterprise-grade governance and flexibility.

AWS created the GenAIIC in 2023 to help customers build real business value from then-new GenAI tools. In July 2025, AWS announced an additional \$100 million investment in the center to create new agentic workflow solutions. The center operates as an investment arm within AWS and is tasked with accelerating adoption of generative AI across industries. AWS told us that IDP emerged as a common, recurring use case in customer engagements, prompting the team to formalize reusable patterns and documentation into a deployable accelerator. That comes as no surprise to Deep Analysis; we covered the rapid rise of Amazon Textract and are familiar with its popularity among IDP end users. The accelerator will build on this solid customer foundation.

The IDP accelerator is available on GitHub with no license fee; customers pay only for the AWS services consumed. It deploys via CloudFormation into a customer's own account, including GovCloud for public sector and FedRAMP-constrained buyers.

Commercially, the accelerator drives AWS revenue indirectly through Bedrock consumption and professional services. Claims processing is currently a popular use case, along with strong public sector demand. The team is exploring industry-vertical packaging, with financial services expected to be a focus.



The Technology

We first reviewed the accelerator in November 2025. Since then, AWS has added an impressive array of functionality and continues to launch near-weekly feature and function updates. Trying to keep up can be overwhelming; indeed, the GenAIIC briefing team encouraged us to use our favorite LLM to summarize the vast amount of documentation on the GitHub page. For this report, we focus on what we consider to be some of the platform's standout features.

The accelerator is a serverless, end-to-end IDP pipeline that includes OCR, classification, extraction, validation, and human review, all controlled by a single configuration artifact. AWS maintains a single main code branch for the accelerator, and customers who stay on it inherit every release – new models, new features, UX improvements – at a near-weekly cadence. The platform has role-based access control (administrator, author, viewer, and reviewer), and a single deployment can run different document use cases side by side. AWS provides managed configurations that improve with each release; customers can clone them and adapt their own custom versions. The configurations are exportable as JSON or YAML.

Model Choice and Management

Every document processing step has a model pick list drawn from Bedrock's full AI roster, including Anthropic, OpenAI, Amazon Nova, Meta Llama, Google Gemma, NVIDIA, and Qwen. When a new frontier model appears on Bedrock, it is available for document processing immediately. Just as notable is what AWS built for the escape hatch: inference Lambda hooks let customers route any step through their own Lambda function – to monitor Bedrock calls, apply inference profiles, or bypass Bedrock entirely for an external model. Custom business logic can likewise be injected before processing and after every pipeline step, keeping the customer code loosely coupled and off the accelerator's source tree. This avoids stranding a customer on a dead-end version.

Agentic Extraction

Extraction runs in two modes. Simple mode performs the entire data extraction in a single inference, something that is increasingly viable for larger documents as million-token context windows become standard. While one could theoretically process any size document with simple mode, experience has proven that the extraction accuracy can seriously degrade as context limits are approached.

For those long and dense documents, advanced mode uses agents for “sharding,” which means the system breaks the document into smaller components (e.g., pages, sections, chunks defined by token limits, or logical units like “each table” or “each form field cluster”), and agents operate separate inference calls on each one. Each inference gets its own smaller, cheaper, or faster model – or no AI at all – rather than an entire document being sent to the LLM in one giant and costly call. A common example is tabular data. For a

50-page document full of dense, high-quality tables, the agent might recognize that the OCR markdown has already created a high-confidence table structure and invoke a deterministic Python table tool instead of an expensive LLM inference.

Schema attributes carry data types, regular-expression patterns, enumerations, defaults, and length constraints; the agent supplies these to the model and verifies the response against them, re-querying with an improved prompt when a value fails. MCP tool support is planned next, which could enable validation of extracted values against external systems of record; the classic example is invoice-to-ERP lookups to catch duplicate or fraudulent invoices. The number of concurrent agents is set at a default of 10. Since this can scale as serverless processes, one could theoretically run hundreds of concurrent agents. However, as the costs can quickly add up, the practical ceiling would be a customer's Bedrock tokens-per-minute quota.

OCR as an Option

In a market where many IDP startups skip the classic OCR step entirely and feed page images straight into vision language models, AWS also offers OCR because the team has learned that downstream data accuracy is often optimized when the model receives both the OCR text and the page image. Customers can select from the following options: Amazon Textract (the default, with tables, layout, signatures, and forms) at roughly five to six cents per page; Bedrock Data Automation as an OCR-only engine at around one cent per page with comparable features; or a vision language model where costs will vary. If test results show a VLM-only pipeline can meet the accuracy and cost parameters, the customer can switch OCR off and stop paying for it.

Confidence Calibration and Human Review

The AWS team told us it is working to provide reliable confidence scores that track accuracy, free of the false positives that have plagued LLMs and flagging for human review only where extraction has genuinely failed. Historically, asking an LLM to extract and self-score in a single inference has produced overconfident and poorly calibrated results; therefore, the accelerator scores confidence in a separate inference. AWS's data science team says this may change with the newest super-large AI models that can calibrate in a single pass, but the trade-off is running up significantly higher model costs.

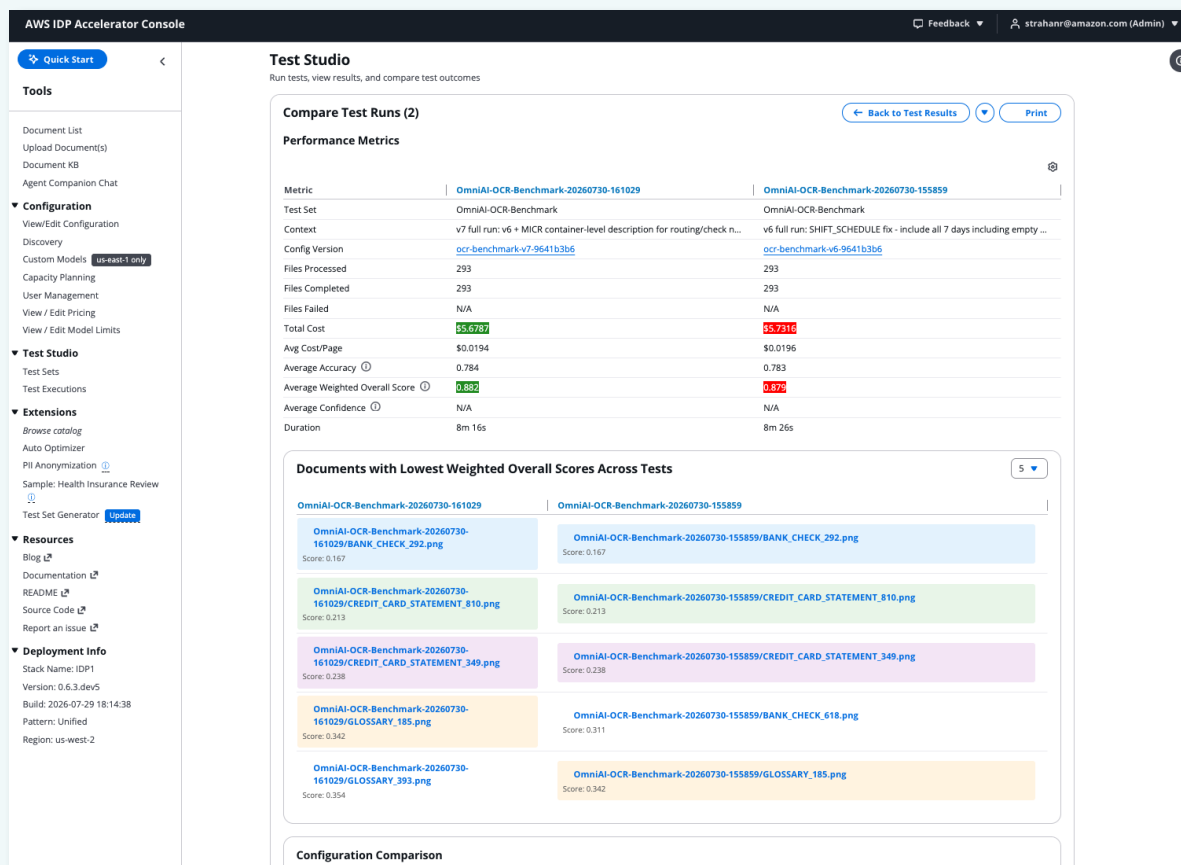
The AWS team suggest that this higher cost could be offset by a meaningful reduction in the false-positive human reviews, which are the most expensive “inference” of all.

Test Studio, Auto Optimizer, and Extensions

Test Studio is the empirical backbone: test sets (documents with ground truth) plus test executions that run a set against any configuration version and compare cost and accuracy, field by field, across models and settings. Four open-source benchmark sets are available on Hugging Face. A free test set generator extension fabricates synthetic documents with known ground truth, including progressive image degradation to mimic poor-quality scanned or faxed documents. Per-document cost transparency and granularity run throughout the process; in the demo, a sample lending package shows how its 19-cent-per-page cost is broken down (see Figure 1).

The Auto Optimizer is an AI agent that understands the accelerator's configuration structure, takes a test set, the budgeted cost

Figure 1
The Test Studio



per page, and a spending cap, accepts natural-language guidance on what “optimal” means, and then iterates configurations against accuracy and cost until they converge (see Figure 2).

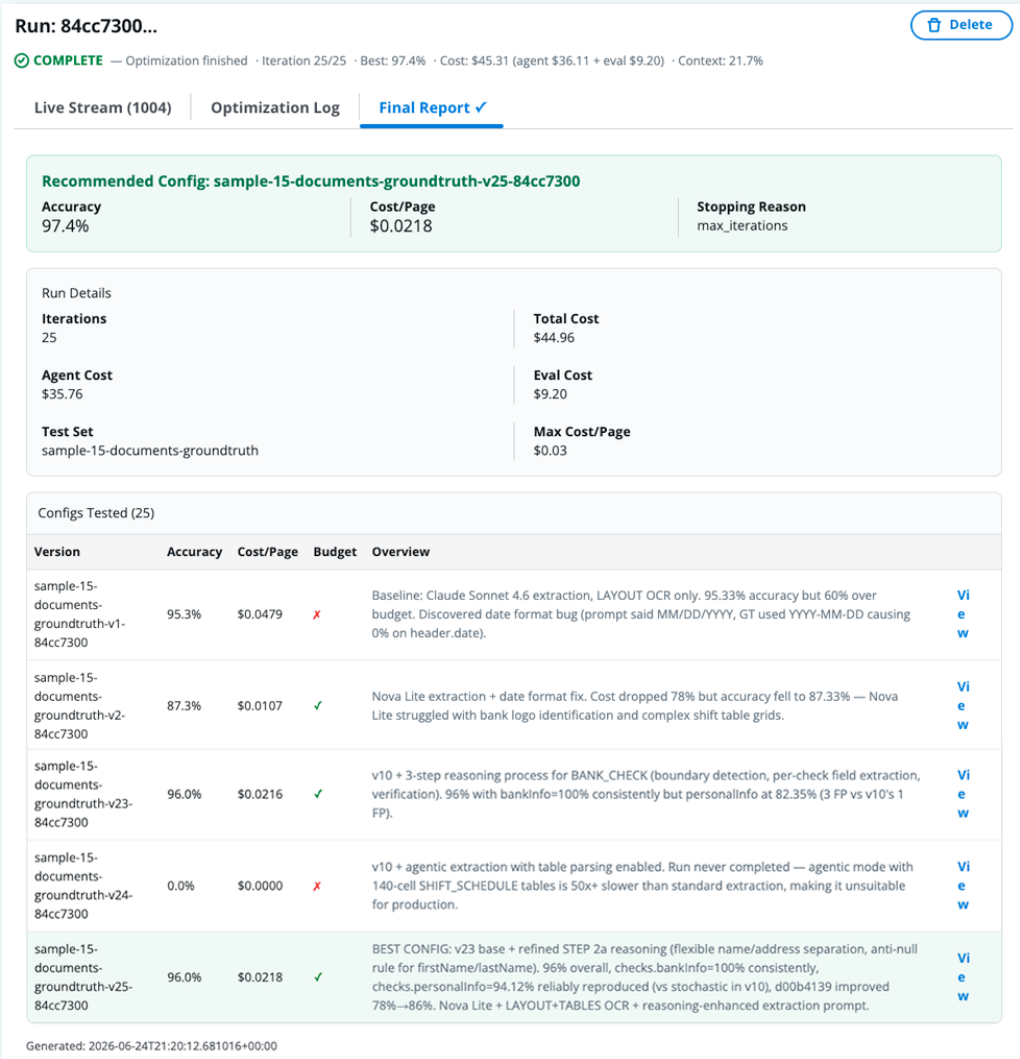
An Agent Companion chat rounds out the platform, integrating DeepWiki’s MCP to answer questions about the system itself, troubleshoot individual document runs, and query extracted data, which is automatically written as Parquet into an Athena SQL database, with the agent composing SQL and generating visualizations on request. In the demo, the AWS team easily generated a distribution chart of income levels based on data extraction results from across thousands of processed US IRS W-2 tax forms.

AWS has also introduced an extensions catalog for accelerator customers to choose from; some are open source and free, while others will be offered as paid subscriptions through AWS Marketplace.

Our Opinion

When we first analyzed the AWS GenAI IDP Accelerator in November 2025, we judged it a capable developers-only offering because of three confusing processing patterns and a steep learning curve. But nine months later, the progress is striking. The addition of quick-start agents, AI-driven auto-configuration, managed configuration versions, and the

Figure 2
The Auto Optimizer



Agent Companion have moved the accelerator toward easier access by a business analyst without dumbing down the deep configurability underneath. We think that layering (sophistication retained, complexity hidden) is the right architecture for a maturing IDP product, and it is precisely what some vendors have gotten wrong by focusing on one direction at the expense of the other.

The cost-and-accuracy optimization feature is particularly noteworthy. We have argued for years that IDP buyers need empirical cost prediction and accuracy measurements before deploying an LLM-powered solution. AWS is ahead of the commercial field here, partly because of its inherent advantage as a cloud hyperscaler that controls every cost and accuracy lever in the pipeline. The new Auto Optimizer extension takes that further and is, in our view, worthy to be sold on its own through the extensions catalog. The confidence-calibration work also addresses one of the longest-standing weaknesses of LLM-based extraction. We also liked the Test Studio and its ability to simulate scanned and fax images as synthetic test documents. If this works as expected, the costly, time-consuming, and frustrating work of collecting sample documents could become obsolete.

The open questions are structural, not technical. A weekly release cadence is a double-edged sword that risks overwhelming all but the savviest users. The emerging Marketplace extension model puts AWS in delicate territory with the IDP ISVs that build on its own services. Providing a free, increasingly comprehensive IDP platform changes the calculus for any vendor whose IDP pipeline is built around Bedrock and Textract.



Advice to Buyers

The accelerator is best suited for organizations already committed to AWS that have engineering capacity and want maximum model choice, full pipeline control, and pay-per-use economics – all without any IDP software licenses. Claims processing and public sector document workloads are proven territory, and AWS GovCloud support covers FedRAMP-bound agencies and contractors. It is not for buyers who need on-premises or non-AWS deployment, nor for teams that want a shrink-wrapped product from a vendor that will own the outcome. This is a build-with-help motion, with AWS professional services and the GenAIIIC team available behind it.

We advise buyers to stay on the main branch and inject your own customization exclusively through the Lambda hook points provided by AWS. Use Test Studio from day one, turn human-reviewed documents into ground-truth baselines, and make configuration changes only against measured cost and accuracy deltas. Rigorously monitor the synthetic document creator against your own ground-truth documents and keep a close watch for variations. Benchmark the Bedrock Data Automation OCR option against Textract early, since the per-page costs could differ significantly. Model the Bedrock tokens-per-minute quota against your peak volumes and make sure to request increased capacity ahead of need. And if you have high-volume, variable document workloads, test the Auto Optimizer. As models continue to churn with no end in sight, continuous re-optimization has become a serious operational problem. While most IDP platform vendors offer some model operations management, this might be the most comprehensive answer to the problem that we have seen to date.

Strengths

- The advantages of being a hyperscaler with a GovCloud contract
- Open source with no license fee; weekly release cadence on a maintained main branch
- Agentic extraction with cost-aware tool selection and constraint-verified retry

Opportunities

- Package managed configurations for industries
- Monetize the Auto Optimizer and future extensions through AWS Marketplace
- Attract partners to build and contribute extensions to the catalog

Aspirations

- Evolve from an engagement accelerator into a platform with an extension ecosystem
- Empower business users to become self-sufficient on the accelerator
- Solve the LLM confidence calibration problem

Results

- Substantial customer adoption and a growing base of customer engagements
- Claims processing established as a flagship use case
- Strong US public sector traction through GovCloud

About Deep Analysis

We Research Innovation



Deep Analysis is an advisory firm that helps technology vendors, buyers, and investors understand and address the challenges of innovative and disruptive technologies in the enterprise software marketplace.

The firm's work is built on decades of experience advising and consulting to global technology firms large and small, from SAP, Oracle, and HP to countless start-ups.

Led by Alan Pelz-Sharpe, Deep Analysis works with technology vendors, buyers of enterprise technology, and investors in the ECM and enterprise automation market to improve their understanding of the information management technology landscape and provide actionable guidance.

Deep Analysis' timely book, "Practical Artificial Intelligence: An Enterprise Playbook," outlines strategies for organizations to avoid pitfalls and successfully deploy AI.

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